SMART PARK: IoT AND ML BASED REAL-TIME PARKING SLOT AVAILABILITY SYSTEM

Suganthen P
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies, Chennai

Sathish kumar K
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies, Chennai

Chandru K
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies, Chennai

Dr. N. Meenakshi
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies, Chennai

Abstract — Parking areas can be hard to use when drivers don't know which spots are open or when reserved spots will be free. This paper introduces SMART PARK, a system that brings together Internet of Things (technology with machine learning to predict space availability. An infrared (IR) sensor is attached to each parking spot. An ESP32 device sends updates via Wi-Fi to Firebase indicating whether the spot is taken. A mobile app shows the status of each spot and allows people to make reservations based on a start date, start time and end time. The app checks existing reservations before confirming one. The times when parking sessions start and end are also saved to create a history that can be looked at later. A Random Forest model uses information such as the day of the week, the time, the weather and past space usage to predict how many spots will be free at a given time. This system brings together checking, planning ahead, booking and predicting all in one process. Testing with a planned prototype is still needed, and possible improvements could include automatic number-plate recognition and more ways to handle reservations.

Key Words — IOT; Smart Parking; ESP32; IR Proximity Sensor; Cloud/Firebase; Real-time Occupancy; Time- Reservation; Parking Session Management; Machine Learning.

INTRODUCTION

In busy parking lots, the challenge of finding a space is a frequent issue, especially as drivers end up spending a lot of time searching for a vacant parking spot. Drivers may have to travel between parking areas in search of an empty space if there is no information on parking availability. This wastes time, carbon emissions and creates unnecessary traffic. Smart parking systems are being designed to overcome these challenges by using Internet of Things (IoT) sensors, wireless communication, cloud computing, mobile applications and machine-learning techniques to enhance the management of parking.

More than just knowledge of available parking, an effective parking management system should offer information about spaces already taken. A slot may be filled now, and open the following time, or a different slot may be full for a particular time period. Without information on the actual or anticipated parking times, and where users will park in the future, it is impossible to reliably schedule parkings. Thus, a smart parking solution must include information on when parking will be available and the ability to reserve it, all in realtime, as well as provide realtime information on the parking space's occupancy.

In order to meet these requirements, SMART PARK will make use of an integrated, IoT and machine-learning-based approach. There is an infrared (IR) sensor for each parking spot to detect the presence of vehicles, and an ESP32 controller is used to transmit collected data to a cloud-based service via Wi-Fi. The mobile application shows the availability of parking spaces in real-time, as well as allows users to choose a parking space and set a time, e.g., 11:00 a.m. to 12:00 p.m. The reservation system is designed to ensure that there are no conflicts in time while the users can plan their parking in accordance with the existing reservations. In addition, a Random Forest model which is trained on historical parking data predicts the availability of the parking slots at a selected time of the next day. SMART PARK will integrate real-time monitoring, advanced booking, parking-session timing and availability prediction into a single system to cut back on users' parking search efforts, optimize the use of parking slots and offer a more convenient and organized parking experience.

We implement Parking Guidance and Information System (PGIS) idea in a campus. An IR sensor is placed in each slot, the ESP32 gathers the data from the sensors and sends it to the cloud via Wi-Fi, and the mobile app displays data stored in the cloud. We wanted these parts to work in concert, and so the sensor data goes into the cloud database, along with the booking logic and the prediction model (not stopping at the detection of the vehicle).

It doesn't really help if someone is going to come in the afternoon when they can see the parking lot live. Therefore, we think it is necessary to implement the time based reservation. The user provides a start time and an end time, the application then determines if that time slot is booked into the system or not. On arrival, the vehicle will be identified by the IR sensor as either being physically occupied or empty. When separation between planned time and actual occupancy are maintained, the application can display the current occupancy and the expected availability.

The prediction section is based on 2 weeks of parking record data. A Random Forest model [12] is built with the information of day, time, weather and occupancy to predict the numbers of free slots in a selected future time. It is designed to be used as a planning tool for those who have yet to leave and will not be used in place of live sensor data.

II. CORE BASE PAPER AND LITERATURE REVIEW

Our base paper is the survey by M. G. Diaz Ogás, R. Fabregat, and S. Aciar, "Survey of Smart Parking Systems," Applied Sciences, vol. 10, no. 11, p. 3872, 2020 [1]. The authors went through 274 publications and sorted smart parking systems by type, by the way vehicles are detected, by the algorithms used, and by the stage of development. Two of the main categories they describe are parking reservation systems (PRS) and parking guidance and information systems (PGIS), and they also look at how sensing and algorithms were combined in different systems [1].

For our project the most useful part is the discussion of open problems: gathering information, user preferences, routing, and rescheduling. The survey argues that a parking system works better when its information is timely and when it can respond as conditions change, and it says that user preferences deserve more attention when guiding people to a space [1]. We treat these points as the motivation for SMART PARK and not as a design to be copied.

Several other papers supplied with the project widen the picture. Channamallu et al. review 124 smart parking studies that cover sensors, wireless links, vision, prediction and user-facing services [2].

Awan et al. compare Decision Tree, Random Forest, KNN, Multilayer Perceptron and ensemble models for predicting parking availability

[3]. Kuhail et al. build a forecasting model from past occupancy and contextual factors [4]. Perković et al. test several parking sensor technologies and discuss reliability and environmental effects, which matter for infrared sensing [5]. Paudel et al. look at demand on a university campus using class schedules, floor-space information and arrival data [6].

A few prototypes are closer to what we are building because they connect hardware to a cloud service and a phone. Elfaki et al. use sensing, an ESP32-CAM with a Raspberry Pi, Firebase and a mobile booking app [7]. Kodali et al. use an ESP32-LoRa gateway and an Android application to display parking status [8]. Our system differs in that it uses one IR sensor per slot, lets users book a time window, and predicts next-day campus availability, all in a single workflow.

III. RESEARCH GAP AND CONTRIBUTION

Across the papers we read, sensing, cloud delivery, prediction and user services are usually treated as separate problems. Many datasets come from cities or from a single site, and a fair number of the proposed solutions are only prototypes or simulations. The base survey adds that information gathering is limited, user preferences are barely used, realtime information is needed, and rescheduling matters when parking conditions change [1].

SMART PARK makes four contributions in response. First, it uses lowcost IR sensors on each slot, with an ESP32 and Wi-Fi, to monitor a campus in real time. Second, it lets the user reserve a start and end time, and the app checks the interval for conflicts. Third, it adds a Random Forest model trained on parking history collected on the campus, to estimate how many slots will be free later. Fourth, it brings live availability, reservation, parking-session timing and prediction together in one mobile app.

TABLE II CORE BASE PAPER GAPS AND SMART PARK RESPONSE

Observed gap / challenge	SMART PARK response
Need for accurate, timely slot information [1]	IR sensor → ESP32 → Wi-Fi → Firebase → mobile live status.
User preferences are weakly represented [1]	User supplies desired date, start time, end time, and block/slot preference.
Need for real-time handling of changing conditions [1]	Cloud state is updated continuously and actual sensor occupancy overrides planned end time.
Research often separates sensing and prediction	SMART PARK combines live IoT sensing with Random Forest future availability prediction.

IV. PROPOSED SYSTEM ARCHITECTURE

The architecture is shown in Fig. 1 and has four layers. The bottom layer detects whether a vehicle is present. Above it, the edge layer gathers the sensor states and sends them on. The cloud layer stores and processes the data, and the application layer turns the results into something the user can act on. We took this layered split from the base survey and fitted it to a campus setting.

SMART PARK: IoT, Cloud, ML, and Mobile Application Architecture

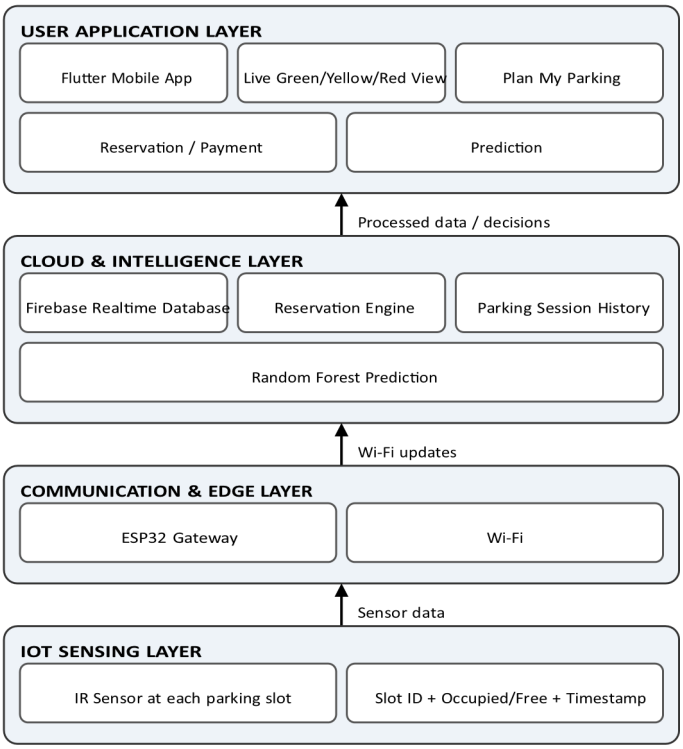


Fig. 1. Proposed SMART PARK architecture: IR sensing, ESP32/WiFi communication, cloud and intelligence services, mobile user application.

V. METHODOLOGY

A. IoT DATA ACQUISITION

Every parking slot has an IR proximity sensor. When its state changes, we read that as a vehicle either arriving or leaving. Sensors can misfire because of noise or a poor mounting position, so the readings can be calibrated and a change can be required to show up in several consecutive readings before it is accepted. If the readings keep disagreeing, the slot is marked unknown and we do not force a free or occupied answer. Each accepted change is saved with the slot identifier and a timestamp, which lets the app show when a parking session began.

For a parking area with N slots, the number of free slots at time t is: $F(t) = N - \Sigma O(t)$ i i where O (t) is 1 if slot i is occupied and 0 if it is not. The count i shown to users is worked out from the latest accepted sensor state
stored in the cloud.

B. EDGE COMMUNICATION AND CLOUD STORAGE

The ESP32 reads the slot sensors and sends the status over Wi-Fi. Firebase holds the current occupancy, the reservation records, the session timestamps and the historical observations. Since many users read and write the same data, keeping it in one cloud database gives the mobile app and the booking logic a single shared source. One record holds the slot ID, block ID, state, timestamp, reservation interval, user ID and session status.

C. TIME-AWARE RESERVATION

Booking by time window is the main feature users interact with. A request carries the date, the slot, a start time and an end time. A plain "book now" button would only look at the present moment, whereas here the whole requested interval is checked against the other bookings for that slot. Let the new interval be [S , E) and an existing one be [S , E). n n e e

The two overlap when:

$$S < E \text{ AND } E > S \; n \; e \; n \; e$$

If this holds, the slot is rejected for that request; if no booking overlaps, the slot is shown as bookable. The app also displays the next reservation on a slot, so users can plan around periods that are occupied or already booked. Once a vehicle has arrived, the IR sensor is what decides physical occupancy, so a booking that has passed its end time does not by itself free the slot.

TABLE III EXAMPLE OF TIME-AWARE SLOT PLANNING

Slot	Existing booking	Requested time	Decision
A05	10:00–12:00	12:30–14:00	ACCEPT
A05	10:00–12:00	11:30–13:00	REJECT
A07	13:00–14:00	11:30–13:00	ACCEPT

D. MACHINE-LEARNING PREDICTION

We developed the model on two weeks of historical parking records. The inputs are the day, the time, the weather and past occupancy, and the quantity to predict is the number of free slots at a chosen future time. We picked Random Forest for the prototype because tree ensembles work well on tabular data and have already been tried for parking availability forecasting [3], [4].

The workflow runs through data collection, cleaning, feature construction, a chronological train/test split, training and evaluation. We split by time and not at random, because a random split lets future records leak into the training set and makes the results look better than they are. The final version should state the number of records, the training and test periods, how the target is defined, which features were used, the error metric and the prediction performance. Our project files do not have those numbers yet, so we do not report any accuracy figure here.

E. MOBILE APPLICATION WORKFLOW

We designed the app so that someone using it for the first time can follow it. The home page shows the number of free and occupied spaces in each parking block. Green, yellow and red indicators show how full a block is, and each slot appears as available, reserved or occupied.

In the “Plan My Parking” screen the user enters a date, a start time and an end time. The app then lists the slots that stay free for the whole interval according to the reservation database. For an occupied slot it can show when the current parking session started, if that is known. A confirmed booking can be linked to the payment layer, and notifications can remind the user when a booking starts and ends. Payment is treated here only as a service layer of the app; the security of the payment gateway and real transaction handling are outside what we evaluate in this paper.

VI. CORE COMPONENTS AND ALGORITHM

The system is made up of the following parts: IR sensors to detect whether a vehicle is present; an ESP32 to collect the readings at the edge; Wi-Fi for communication; Firebase for live state, reservations, sessions and history; the mobile app for monitoring, planning, booking and notifications; and a Random Forest model for predicting future free slots.

The booking procedure goes as follows:

1) Receive date, start time, end time, and preferred block/slot.
2) Read the latest slot state from the cloud database.
3) Retrieve existing reservations for the requested slot and date.4) Reject any candidate where $S < E$ AND $E > S$. n e n e
5) Show the slots without conflicts, along with current and nextavailability.
6) When the user confirms, create the reservation and thepayment request that goes with it.
7) On arrival, check actual occupancy with the slot's IR sensor.
8) Update the session timestamps and keep the slot unavailablewhile it is occupied.

VII. SYSTEM VALIDATION AND RESULTS PLAN

At this point we have the architecture and the plan for the two-week dataset, but no measured results from the prototype. This paper therefore describes the design and how we intend to validate it, and it makes no performance claims that we cannot support.

Once the system is built, we plan to evaluate it on four fronts: sensor correctness, communication latency, prediction quality and how well the app works for users.

TABLE V RECOMMENDED FINAL EVALUATION METRICS

Metric	Measurement
Sensor accuracy	Correct occupied/free states against manually labeled ground truth.
Update latency	Time from sensor state change to cloud and mobile update.
Prediction error	MAE or RMSE between predicted and actual free-slot counts.
Booking conflict rate	Percentage of overlapping requests incorrectly accepted.
Search-time change	Difference in average parking-search duration with and without the app.

Each of these ties back to a goal of the project. The booking-conflict test checks that the reservation logic does what it should, and comparing search times with and without the app would give real evidence that planning becomes easier. After the prototype has been tested, the measured values will replace these placeholders in the camera-ready version.

VIII. ADVANTAGES AND APPLICATIONS

We see several practical benefits for campus parking. Live slot information removes some of the guesswork about what is free right now. Time-based reservation lets people plan around slots that are already booked, so they are not simply turning up and hoping. Session timestamps show how long a vehicle has been in a slot, within the limits of what the sensor can detect. The prediction gives one more hint when someone is deciding what time to come. Because the design is cloud-based, it can be extended to more parking blocks and more kinds of users.

The same approach could be used in universities and colleges, office campuses, hospitals, shopping facilities and other controlled parking areas where slot-level sensing and advance booking make sense.

IX. LIMITATIONS AND FUTURE ENHANCEMENTS

The prototype has some clear limitations. Infrared sensing can be affected by where the sensor is placed, by the shape of the vehicle and by the environment [5]. The machine-learning dataset covers only two weeks in our current plan, so we still need to check how well the predictions hold up over longer periods. The system is a prototype and has not been tried at large scale. Reservations also depend on people using the app consistently, and on clear rules for late arrivals, cancellations and vehicles that stay after the booked end time.

Later work could collect data over several semesters, calibrate the sensors automatically, and add inputs such as academic schedules and campus events. Other ideas include dynamic pricing, automatic

number-plate recognition, electric-vehicle charging, better navigation and deployment across several campuses. A production payment system would also need defined rules for authentication, transaction security, cancellation, refunds and failure recovery.

X. CONCLUSION

This paper presented SMART PARK, a smart parking framework for campuses that combines IoT sensing, cloud synchronization, timebased reservation, a mobile app and machine-learning prediction. The survey of 274 smart parking studies that we used as our base paper supplied the main motivation, since it points to problems with information gathering, user preferences, real-time processing and rescheduling [1]. We turned these points into a working flow: IR sensors report the physical occupancy, the ESP32 and Wi-Fi carry each change to the cloud, the cloud keeps the shared parking data, the app handles planning and booking, and a Random Forest model estimates future free slots from past data. One principle runs through the design: the reservation describes what a user intends to do, while the sensor describes what is actually happening. That makes the system easier to use for anyone planning a parking interval, and safer when real parking behavior does not match the booking. Our next step is to finish deploying the prototype and to replace the planned metrics with measured results before the final submission.

REFERENCES

[1] M. G. Diaz Ogás, R. Fabregat, and S. Aciar, "Survey of Smart ParkingSystems," Applied Sciences, vol. 10, no. 11, p. 3872, 2020, doi: 10.3390/app10113872.

[2] S. S. Channamallu, S. Kermanshachi, J. M. Rosenberger, and A.Pamidimukkala, "A Review of Smart Parking Systems," Transportation

Research Procedia, vol. 73, pp. 289–296, 2023, doi: 10.1016/j.trpro.2023.11.920.

[3] F. M. Awan, Y. Saleem, R. Minerva, and N. Crespi, "A ComparativeAnalysis of Machine/Deep Learning Models for Parking Space Availability Prediction," Sensors, vol. 20, no. 1, p. 322, 2020, doi: 10.3390/s20010322.

[4] M. A. Kuhail, M. Boorlu, N. Padarthi, and C. Rottinghaus, "ParkingAvailability Forecasting Model," in Proc. IEEE International Smart Cities Conference (ISC2), Casablanca, Morocco, 2019, pp. 619–625, doi: 10.1109/ISC246665.2019.9071688.

[5] T. Perković, P. Šolić, H. Zargariasl, D. Čoko, and J. J. P. C. Rodrigues,"Smart Parking Sensors: State of the Art and Performance Evaluation,"

Journal of Cleaner Production, vol. 262, p. 121181, 2020, doi: 10.1016/j.jclepro.2020.121181.

[6] S. Paudel, M. Vechione, and O. Gurbuz, "Predicting University CampusParking Demand Using Machine Learning Models," Transportation Research Record, vol. 2678, no. 6, pp. 14–26, 2024, doi: 10.1177/03611981231193417.

[7] A. O. Elfaki, W. Messoudi, and A. Bushnag, "A Smart Real-Time ParkingControl and Monitoring System," Sensors, vol. 23, no. 24, p. 9741, 2023, doi: 10.3390/s23249741.

[8] R. K. Kodali, K. Y. Borra, S. S. G. N. Sharan, and H. J. Domma, "An IoT

Based Smart Parking System Using LoRa," in Proc. 2018 International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery (CyberC), Zhengzhou, China, 2018, doi:

10.1109/CyberC.2018.00039.

[9] E. I. Vlahogianni, K. Kepaptsoglou, V. Tsetsos, and M. G. Karlaftis, "AReal-Time Parking Prediction System for Smart Cities," Journal of Intelligent Transportation Systems, vol. 20, no. 2, pp. 192–204, 2016, doi: 10.1080/15472450.2015.1037955.

[10] J. Arjona Martínez, M. P. Linares, J. Casanovas-Garcia, and J. J. Vázquez, "Improving Parking Availability Information Using Deep Learning Techniques," Transportation Research Procedia, vol. 47, pp. 385–392, 2020, doi: 10.1016/j.trpro.2020.03.113.

[11] F. Al-Turjman and A. Malekloo, "Smart Parking in IoT-Enabled Cities: ASurvey," Sustainable Cities and Society, vol. 49, p. 101608, 2019, doi:

10.1016/j.scs.2019.101608.